

Project Intelligence Hub

Comprehensive App Guide

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Live website: <https://samcoegyptdashboard.vercel.app>

Figma reference design: <https://www.figma.com/design/z1T4ERNeuBlrpC3slf93KT>

1. Executive Summary

Project Intelligence Hub is a project-controls, delay-analysis, claims-intelligence, and executive-reporting system. It currently has two working surfaces:

- Original Streamlit app: the full Python application, advanced calculations, uploads, exports, and detailed project-controls workflows.
- Next.js/Vercel website: public, fast, mobile-friendly executive website generated from the same project folders and HTML outputs.

The main operating rule is project isolation: each selected project reads from its own project folder, shows its own data, and writes or publishes its own outputs.

2. Current Portfolio Snapshot

- Projects detected: 4
- Sectors detected: 3
- Total contract value: EGP 4,200,000,000
- Average progress: 100.0%
- Average SPI: 0.75
- Average CPI: 60.11
- Decisions required: 1

Project	Sector	Status	Contract Value	Progress	Activities	Decision Required
LMD Bridge & Road Interchange	Bridges	Delayed	EGP 850,000,000	100.0%	17	No
ROYA-BIG PROJECT PHASE01 (B1-4)	Buildings	Active	EGP 0	100.0%	1363	Yes
Sophia Mall Mixed-Use Development	Buildings	On Track	EGP 2,100,000,000	100.0%	17	No
Suez Tunnel Civil & MEP Works	Tunnels	Delayed	EGP 1,250,000,000	100.0%	17	No

3. Sector Structure

Projects are grouped by sector folders under `projects`. Adding a new sector folder automatically creates a new sector in the Decision Making Dashboard after the generator runs.

Sector	Projects	Contract Value	Progress	SPI	CPI	Decisions
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Bridges	1	EGP 850,000,000	100.0%	1.0	78.3	0
Buildings	2	EGP 2,100,000,000	100.0%	0.5	41.92	1
Tunnels	1	EGP 1,250,000,000	100.0%	1.0	78.3	0

4. High-Level Architecture

```
flowchart LR
    A["Project folders"] --> B["Python data generator"]
    B --> C["website/public/data JSON"]
    B --> D["website/public/generated HTML outputs"]
    C --> E["Next.js website"]
    D --> E
    E --> F["Vercel public URL"]
    A --> G["Streamlit original app"]
    G --> H["Advanced reports, TIA, claims, exports"]
    I["RUN_FULL_PROJECT_NO_GIT_SYNC.bat"] --> J["GitHub repository"]
    J --> F
```

5. Main User Flow

1. Open the website.
2. Decision Making Dashboard appears first.
3. Choose a project from the Active Project dropdown.
4. The same page updates the Project Workspace tabs.
5. Use the old-app-style tabs for Overview, EBS/WBS, Activities, Milestones, S-Curve, EVM, Contracts, Risk Matrix, Letters Intelligence, Delay Analysis, Contract & Claims, Conference, and Output Studio.
6. Update source files in the selected project folder.
7. Run the website data generator and sync.
8. Vercel updates the public dashboard from GitHub.

6. App Tabs and Features

Tab / Page	Purpose	Main Data or Behavior
Decision Making Dashboard	Top-management portfolio view	Aggregates all project JSON into portfolio KPIs, sector view, project console, and report viewer.
Overview	Project executive summary	Shows project status, value, progress, remaining value, source metadata, and overview table.
EBS / WBS	Work breakdown view	Shows WBS source table and master dashboard embed for the selected project.
Activities	Activity register	Shows activity, progress, EVM, and delay event counts with source-table preview.
Milestones	Milestone tracking	Shows milestone records, SPI schedule health, delay days, and milestone source table.
S-Curve	Progress trend	Shows S-curve source status and linked executive dashboard HTML.
EVM	Earned value management	Calculates BAC, PV, EV, AC, SPI, CPI from selected project data.
Contracts	Commercial position	Shows contract value, paid, spent, remaining value, contracts and payments previews.
Risk Matrix	Risk and decision signals	Shows risk score, risk rows, decision trigger, delay exposure, and risk table.
Letters Intelligence	Correspondence intelligence	Detects inbox files and workbook, shows letter inventory and generated dashboard output.
Delay Analysis	Time Impact Analysis	Detects TIA templates, missing files, column previews, MEP schedule, BL schedule, and generated delay charts.
Contract & Claims	Claims knowledge base	Detects contract files, clause library, SQLite tables, evidence, and claim exposure.

Conference	Meeting support	Reads meeting_url from project.json and provides same-page conference panel and join button.
Output Studio	Generated reports	Shows four automatic HTML outputs and watcher/sync status for the selected project.

7. Folder Guide

Folder / File Area	Purpose	How It Is Used
projects/{Sector}/{Project}	Main project workspace	Add future projects here under sector folders. The generator detects them automatically.
01-data/import_templates	Core project data	CSV files for projects, activities, progress, EVM, payments, contracts, claims, risks, milestones, WBS, S-curve.
02-delay_analysis	TIA workspace	Methodology documents and steel_delay_tia_templates CSVs used by Delay Analysis.
03-schedule	Schedule support	BL Schedule, MEP Schedule, MEP Activities, and civil logic files.
04-source_excel	Raw Excel source	Original Excel files before conversion into controlled CSV inputs.
05-contracts	Contract intelligence	Contract source PDFs, clause libraries, contract_claims.db, and clauses.
06-evidence	Evidence register	Evidence templates, photo logs, document references, and claim evidence.
07-letters_intelligence	Correspondence intelligence	letters_intelligence.xlsx plus inbox folders for new letters.
08-branding	Project branding	Logo placeholder, identity, palette, and report branding templates.
09-notes	Project notes	Meeting, engineering, and claims notes.
10-deliverables	Formal deliverables	Generated project deliverables outside the automatic website HTML output set.
11-outputs	Automatic HTML outputs	Per-project generated HTML outputs: executive, master, SVG charts, linked dashboard.
12-logs	Project logs	Project-specific operational logs.
website	Next.js public site	Vercel website source, React UI, CSS, and public data.
tools	Automation tools	Data generator, GitHub no-Git sync, validation, lineage, and package builders.
src/construction_system	Python backend logic	Streamlit support modules, project context, loaders, TIA logic, reporting, OpenAI gateway.
dashboard.py	Original Streamlit app	The full original application with Python-based calculations and advanced workflows.

8. Project Folder Standard

Every project should live at:

``projects/{Sector Name}/{Project Folder Name}``

Minimum files:

- ``project.json``
- ``project_manifest.json``
- ``01-data/import_templates/*.csv``
- ``02-delay_analysis/steel_delay_tia_templates/*.csv``
- ``03-schedule/*.csv``
- ``05-contracts/source/*``
- ``07-letters_intelligence/inbox/*``

Recommended ``project.json`` fields:

```
{
  "project_id": "unique-project-id",
  "project_name": "Visible Project Name",
  "client_name": "Client",
  "contractor": "SAMCO",
  "currency": "EGP",
  "status": "Active",
  "meeting_url": "https://your-meeting-link"
}
```

9. Data Pipeline

The website data pipeline is:

Project folders -> `tools/generate\_nextjs\_website\_data.py` -> `website/public/data/portfolio.json` and `website/public/data/projects/\*.json` -> Next.js UI.

The generator also copies automatic HTML outputs from root `11-outputs/{Project Folder Name}` into `website/public/generated/{Project Folder Name Slug}`.

10. Automatic HTML Outputs

Each project should have exactly these automatic website HTML outputs:

File	Name	Purpose
01_executive_dashboard.html	Executive dashboard	Project-level management dashboard HTML.
02_master_dashboard.html	Master dashboard	Detailed project dashboard by sections.
03_elite_svg_charts.html	Elite SVG charts	Tabbed SVG chart gallery generated as HTML.
04_linked_executive_dashboard.html	Linked executive dashboard	Linked A3-style executive dashboard HTML.

11. Letters Intelligence

Letters Intelligence is project-scoped. The website detects:

- `07-letters\_intelligence/letters\_intelligence.xlsx`
- `07-letters\_intelligence/inbox/\*\*`
- letter file count, extension, size, modified date
- linked claims and delay-event availability

To add new letters, place them in the selected project inbox folder, then regenerate and sync.

12. Delay Analysis - Time Impact Analysis

Delay Analysis reads project-specific TIA templates from:

`02-delay\_analysis/steel\_delay\_tia\_templates`

The website detects the file inventory, row count, column count, missing required files, MEP activities, MEP schedule, MEP civil logic, and BL schedule. The original Streamlit app remains the full calculation engine for deep TIA exports and report generation.

13. Contract & Claims Intelligence

Contract & Claims uses:

- `05-contracts/source`
- `05-contracts/source/Overall\_Contract\_clause\_library.xlsx`

- ``05-contracts/contract_claims.db``
- ``06-evidence``

The website shows file inventory and knowledge-base table counts. The original Streamlit app remains the advanced claims engine for extraction, evidence mapping, rebuttal, claim drafting, and exports.

14. Conference Call

The Conference tab reads ``meeting_url`` from the selected project's ``project.json``.

If the field is missing, the tab shows setup instructions. If the field exists, it shows a Join Conference button.

Teams and Zoom usually block iframe embedding, so the meeting may open in a new tab while the dashboard remains available.

15. Output Studio

Output Studio shows the four automatic HTML reports and sync/watch status. Manual Streamlit output features still exist in the original app. The Next.js website focuses on fast HTML report viewing and mobile-safe access.

16. Sync and GitHub

The no-Git sync is controlled by:

- ``RUN_FULL_PROJECT_NO_GIT_SYNC.bat``
- ``tools/github_no_git_sync.ps1``
- ``tools/github_sync_config.json``

Repository target:

- Owner: ahmedlabib33-boop
- Repository: Samco-Project-Intelegant-Hub
- Branch: main
- Default interval seconds: 30
- Deletion sync: True

Credentials must come from ``GITHUB_TOKEN`` or ``GH_TOKEN`` in the local Windows process.

17. Vercel Deployment

Recommended Vercel settings:

- Root directory: ``website``
- Framework: Next.js
- Build command: ``npm run build``
- Install command: ``npm install``
- Output directory: ``.next``

Production URL:

``https://samcoegyptdashboard.vercel.app``

18. Common Commands

Action	Command
Generate website data	cd "D:\Project Intelligence Hub NextJS" && python tools\generate_nextjs_website_data.py
Run website locally	cd "D:\Project Intelligence Hub NextJS\website" && npm run dev
Build website	cd "D:\Project Intelligence Hub NextJS\website" && npm run build
Deploy to Vercel	cd "D:\Project Intelligence Hub NextJS\website" && npx vercel@latest deploy --prod --yes --project samcoegyptdashboard
Sync to GitHub once	cd "D:\Project Intelligence Hub NextJS" && cmd /c RUN_FULL_PROJECT_NO_GIT_SYNC.bat Once 30
Start sync watcher	cd "D:\Project Intelligence Hub NextJS" && cmd /c RUN_FULL_PROJECT_NO_GIT_SYNC.bat Watch 30

19. Adding a New Project

1. Create a sector folder under `projects` if needed.
2. Copy `\_PROJECT\_TEMPLATE` into the sector folder.
3. Rename the copied folder to the project name.
4. Edit `project.json` and `project\_manifest.json`.
5. Add project CSVs under `01-data/import\_templates`.
6. Add TIA files under `02-delay\_analysis/steel\_delay\_tia\_templates`.
7. Add schedule support under `03-schedule`.
8. Add contracts under `05-contracts/source`.
9. Add letters under `07-letters\_intelligence/inbox`.
10. Run the generator.
11. Build and sync.

20. Update Workflow

When project files change:

1. Update the files in the correct project folder.
2. Run `python tools\generate\_nextjs\_website\_data.py`.
3. Run `npm run build` from `website`.
4. Run `RUN\_FULL\_PROJECT\_NO\_GIT\_SYNC.bat Once 30`.
5. Vercel rebuilds from GitHub.

21. Troubleshooting

Issue	Likely Cause	Fix
Project not visible	Folder not under sector folder or generator not run	Place folder under `projects/{Sector}` and rerun generator
Website shows old data	JSON not regenerated or GitHub/Vercel not updated	Run generator, sync, redeploy
Conference button missing	`meeting_url` missing in `project.json`	Add meeting URL and regenerate
Letters not detected	Files not in project inbox	Add files under `07-letters_intelligence/inbox`
Delay tab missing files	Required TIA CSVs absent or named differently	Add files to `02-delay_analysis/steel_delay_tia_templates`
Claims data looks empty	Contract source or DB missing	Add contract source files and rebuild contract library in Streamlit
Sync fails 401	Token not available in process scope	Set `GITHUB_TOKEN` or `GH_TOKEN` in the active Windows environment
Vercel build warning about audit	Dev-package audit noise	Check production audit with `npm audit --omit=dev`

22. Maintenance Rules

- Do not mix project data between folders.
- Do not hardcode project names in the website UI.
- Use `project.json` for project identity and meeting links.
- Use CSV files as the source of visible project data.
- Regenerate website JSON after source changes.
- Keep automatic outputs as HTML in per-project output folders.
- Keep credentials outside the repository.
- Use the Streamlit app for deep Python workflows and the Next.js site for public/mobile viewing.

23. Roadmap

Recommended next improvements:

1. Add a backend API if live online uploads and AI extraction are required on Vercel.
2. Add project-level authentication if dashboard access must be private.
3. Add scheduled GitHub/Vercel automation for automatic rebuilds.
4. Add Playwright tests for tab clicks and mobile rendering.
5. Add a small admin page to edit `project.json` safely.
6. Add export status badges for each generated report.